

From Aggregate Scores to Projectible Profiles: Robustness in AGI Evaluation

Brett Reynolds *

Humber Polytechnic & University of Toronto

Abstract

Contemporary evaluations of artificial general intelligence report broad capability profiles, but their aggregate scores are often asked to support inferences they haven't been validated to support. Stable domain means can conceal offsetting item-level improvements and degradations; stable profile shape can coexist with severe harmful-tail failures; and behavioural stability alone doesn't identify the mechanisms responsible for it. I call a score **PROJECTIBLE** (able to support a declared inference beyond its own construction) and make that property the organizing principle for AGI evaluation. An evaluation should begin with a declared prediction or decision, population, unit of analysis, intervention range, and time horizon. It should then report robustness at four distinct granularities: profile-shape similarity, signed level change, mean absolute item instability, and worst-tail degradation. Retention, response to feedback, and scaffold dependence require separate protocols rather than a universal stability index. Alternative domain weights should likewise earn their use through held-out predictive validity and calibration rather than through claims about which abilities are constitutive of intelligence. The framework preserves the breadth of multidomain batteries while making cancellation, tail risk, transport failure, and evidential limits visible. It yields black-box protocols that are operational now without treating their outputs as evidence of a homeostatic mechanism or a natural kind.

I WHAT SHOULD AN AGI EVALUATION LET US INFER?

Hendrycks et al. (2025) move AGI evaluation beyond isolated benchmarks by adapting Cattell–Horn–Carroll (CHC) theory to a ten-domain capability profile. This broad profile improves on a single leaderboard score. The measurement-theoretic account in Hernández-Orallo (2017) likewise emphasizes explicitness about the abilities, populations, and uses for which scores are interpreted. But the multidomain proposal leaves a prior question open: what should passing the evaluation let us infer?

A score can summarize observed performance without predicting anything beyond the observations used to compute it. Evaluation becomes decision-relevant only when the score travels: from

*I used ChatGPT 5, Claude Sonnet 4.5, and Gemini Pro 2.5 extensively in drafting and revising the paper. I reviewed, edited, and approved all the material and take full responsibility for the final text and conclusions. brett.reynolds@humber.ca

Table 1: Examples of declared projection targets.

Target	Unit	Relevant variation	External outcome
Context robustness	Deployed agent	Irrelevant context type and length	Correctness and harmful failures
Retention	Stateful system	Clean-session delays	Performance on taught, held-out items
Scaffold dependence	Declared system configuration	Tool, retrieval, and context degradation	Task success under component change
Deployment reliability	Deployed system	Users, tasks, and environments	Calibration, completion, and incident rates

tested items to untested items, from one context to another, from a model in the laboratory to a deployed system, or from present performance to later behaviour. I use PROJECTIBILITY for this ability to support a specified inference beyond the score’s own construction (Boyd, 1991; Reynolds, 2026).

Projectibility isn’t a generic property of a metric. It’s always projectibility for a declared target. A release team may need to predict harmful failures under long, noisy contexts. A procurement team may need to predict task completion in a fixed tool environment. A cognitive scientist may ask whether two systems share a representation-based competence. These aren’t interchangeable questions, and a score that works for one needn’t work for the others.

Before choosing metrics or weights, an evaluation should declare five parameters:

1. Projection target: the outcome, inference, explanation, or decision the evaluation is meant to support
2. Population: the model families, versions, deployment settings, and users to which the claim is intended to apply
3. Unit of analysis: a base model, a stateful agent, or a deployed model–tool–retrieval system
4. Intervention range: the contexts, delays, distribution shifts, or component changes across which the projection is expected to hold
5. Time horizon: immediate performance, retention across sessions, or downstream behaviour over weeks or months

Table 1 illustrates how the same broad evaluation can serve different targets only by changing its unit, perturbations, and outcome data.

The field selects which question matters; it doesn’t determine whether the projection succeeds. Once a target is declared, the evaluation has to demonstrate out-of-sample validity, calibration, and transport across the claimed range. This keeps purpose-indexing from becoming conventionalism.

The proposal developed below has four rules. Declare the projection before choosing the score. Keep profile, level, item, and tail granularities separate. Compare every richer model against a simple baseline. Aggregate only when a shared target validates the aggregation. These rules don’t define what AGI is. They set the evidential conditions under which an AGI evaluation can support claims about systems beyond the test itself.

2 WHY AGGREGATE STABILITY ISN'T RELIABILITY

2.1 CANCELLATION AT THE ITEM LEVEL

Suppose a model's accuracy is 70% both with and without a perturbation. It's tempting to call the model robust. The identical means are also compatible with a different result: some questions become much easier, others much harder, and the changes cancel.

Zhang et al. (2026) demonstrate this pattern under task-irrelevant context. Across models and benchmarks, aggregate accuracy often changes little even though individual examples undergo large shifts in both directions. Noise-floor-adjusted item instability (INS) reaches 13.6% across the reported model-benchmark cells, and adjusted worst-decile degradation (10%-WTD) reaches 53.2%. The examples affected are largely model-specific. On MMLU-Pro, cross-model correlations in per-example change average approximately zero.

This isn't merely another prompt-sensitivity result. It identifies a structural defect in aggregate robustness claims. A signed mean can be close to zero because nothing changed or because large positive and negative changes offset. Those states warrant different reliability claims; whether the shifts predict failures in deployments with irreversible actions or along long agent trajectories requires external validation.

2.2 THE SAME PROBLEM RECURS ABOVE ITEMS

Aggregation can hide heterogeneity at several levels. A stable mean within a domain can conceal item-level churn. A stable ten-domain profile can conceal a narrow construction or subprocess that fails sharply. A stable average across models can conceal family-specific failure sets.

The issue already appears in linguistic evaluation. von der Malsburg and Padó (2026) find that transformer models can look human-aligned for one agreement-attraction configuration while diverging sharply in another, warning against projection from isolated configurations or individual models. Kuribayashi et al. (2026) likewise find that layer-specific surprisal from earlier and later layers aligns best with different aspects of human sentence processing; no single layer-specific measure was optimal across both the naturalistic and ambiguity-resolution settings. Neither study tests aggregate AGI scores directly. They provide analogous evidence that the granularity of a score has to match the granularity of the intended projection.

The current lesson for AGI evaluation is direct. A stable domain vector is evidence about a domain vector. It isn't automatically evidence that the same items remain reliable, that the harmful tail is controlled, or that the result transports to another model family.

2.3 STABILITY DOESN'T IDENTIFY ITS CAUSE

A second inference also needs blocking. Observed stability isn't by itself evidence of a mechanism that maintains a capability cluster. Stability, network order, maintenance, and corrective control are separate achievements, and only the last warrants strict homeostatic language (Reynolds, 2026). Weinberger (2026) shows why control is more than constancy: lower-scale adjustments can produce a stable higher-scale control relation. A repeated black-box score establishes neither that relation nor the mechanism producing it.

This distinction matters even if one has no interest in natural-kind theory. A robust behavioural regularity can be useful for prediction while its internal basis remains unknown. Conversely, an elegant mechanistic story doesn't rescue a score that fails to predict its declared target. Projectibility is

Table 2: The proposed robustness outputs.

Output	Granularity	Question
PSS_p	Domain profile	Did relative domain shape persist?
L_{gp}, L_p	Domain and overall level	Did performance rise or fall?
INS_{gp}	Items within a cell	How much two-sided item change occurred?
$WTD_{q,gp}$	Harmful item tail	How severe was the worst q -fraction?
DR_d	Retention at delay d	How much initial performance was retained?
G, B	Feedback trajectory	Did errors decline, and how much backsliding occurred?

the evaluative target; causal evidence can explain or improve it, but it can't be read off behavioural stability.

2.4 EQUAL WEIGHTING IS A BASELINE, NOT A METAPHYSICAL ERROR

The same discipline applies to domain weights. Human CHC research finds differential loadings across abilities (Carroll, 1993; McGrew, 2009; Schneider & McGrew, 2018). Those results don't by themselves establish causal importance in AI systems or decision relevance for every deployment. Transport from human factor structure to machine evaluation is an empirical hypothesis.

Equal weights are a useful baseline: simple, transparent, and easy to falsify. An alternative weighting scheme earns its place only if it improves prediction or decision quality for a declared target on held-out systems. Human loadings, structural hypotheses, or causal interventions may inform candidate weights, but no fixed weighting is a defensible default before that comparison.

3 A MULTI-GRANULAR ROBUSTNESS PROFILE

3.1 SETUP AND NOTATION

Let domains be indexed by $g = 1, \dots, G$, items within domain g by $i = 1, \dots, N_g$, and perturbations by $p = 1, \dots, P$. The baseline condition is $p = 0$. Let

$$s_{igp} = \Pr(A_{igp} = 1)$$

be the latent probability that the system receives credit on item i in domain g under condition p . Repeated responses or probability scores provide estimates of s_{igp} . Define the paired item-level change

$$\delta_{igp} = s_{igp} - s_{ig0}.$$

The corresponding domain score is

$$a_{gp} = \frac{1}{N_g} \sum_{i=1}^{N_g} s_{igp},$$

and $\mathbf{a}_p = (a_{1p}, \dots, a_{Gp})$ is the domain profile under perturbation p .

Table 2 summarizes the principal outputs. They answer different questions and shouldn't be collapsed by default.

3.2 PROFILE-SHAPE SIMILARITY

For each perturbation, compute the correlation between the baseline and perturbed domain profiles:

$$r_p = \text{corr}_g(a_0, a_p).$$

The correlation can be reported directly. If a unit-interval display is operationally useful, define

$$\text{PSS}_p = \frac{1 + r_p}{2}.$$

This mapping changes presentation, not evidential content. A high PSS_p means that relative domain strengths and weaknesses were similar under perturbation p . It doesn't mean that absolute level, individual items, or the harmful tail remained stable.

Uncertainty in PSS_p should propagate score estimation within domains, for example by resampling items and repeated responses within each domain rather than treating the estimated domain means as fixed.

PSS_p is undefined when either profile has zero variance. Pearson correlation is also only one similarity choice. Spearman correlation can test rank preservation, while cosine similarity emphasizes direction. The measure and its interpretation should be preregistered. Disagreement among them is substantive information rather than a reason to pick the most favourable result.

Designed perturbations needn't be exchangeable draws from one common population. Temporal delay, context contamination, scaffold removal, and distribution shift can produce different effects for different reasons. Report PSS_p by perturbation rather than Fisher-pooling heterogeneous interventions by default. A summary across perturbations is permissible only with declared weights representing a specified deployment mixture.

3.3 SIGNED LEVEL CHANGE

For each domain and perturbation, report

$$L_{gp} = a_{gp} - a_{g0},$$

and report the unweighted overall change

$$L_p = \frac{1}{G} \sum_{g=1}^G L_{gp}.$$

Because scores are normalized to $[0, 1]$, these differences lie in $[-1, 1]$. Positive and negative changes remain visible. This replaces a capped ratio such as $\min(1, a_{gp}/a_{g0})$, which discards improvements and can hide offsetting changes when it's the only reported summary.

If target-specific domain weights $w_g^{(Y)}$ have been validated for outcome Y , a weighted change may also be reported:

$$L_p^{(Y)} = \sum_{g=1}^G w_g^{(Y)} (a_{gp} - a_{g0}), \quad \sum_g w_g^{(Y)} = 1.$$

The superscript matters. There's no purpose-free weighted level change.

3.4 ITEM INSTABILITY AND HARMFUL TAILS

Signed level change still permits cancellation within a domain. Adapting the item-level estimands of Zhang et al. (2026), define mean absolute item instability for each perturbation-by-domain cell:

$$\text{INS}_{gp} = \frac{1}{N_g} \sum_{i=1}^{N_g} |\delta_{igp}|. \quad (1)$$

This quantity is in $[0, 1]$. It measures the magnitude of two-sided change, not its direction. Report it beside L_{gp} , since two systems can have the same INS_{gp} while one mostly improves and the other mostly degrades.

To isolate severe degradation, choose a tail fraction $q \in (0, 1]$ before inspecting the results. Let $m_g = \lceil qN_g \rceil$, and let

$$\delta_{(1)gp} \leq \delta_{(2)gp} \leq \dots \leq \delta_{(N_g)gp}$$

be the ordered item changes. Define

$$\text{WTD}_{q,gp} = -\frac{1}{m_g} \sum_{\ell=1}^{m_g} \delta_{(\ell)gp}. \quad (2)$$

A larger positive value indicates worse degradation in the selected tail. If every item improves, $\text{WTD}_{q,gp}$ can be negative; it shouldn't be clamped to zero.

The full distribution of δ_{igp} should accompany these summaries. For binary scoring, also report the estimated mass of correct-to-incorrect and incorrect-to-correct transitions. For probabilistic scoring, report the corresponding directional changes in correctness probability. This makes the cancellation pattern inspectable.

3.5 REPEATED SAMPLING AND NOISE CORRECTION

Neither INS nor WTD can be estimated responsibly from one stochastic response per item. Absolute values create a positive sampling floor even under a true null effect, and selecting the worst observed tail selects negative noise.

Follow the design of Zhang et al. (2026) by collecting repeated, balanced samples under baseline and perturbed conditions. Construct pseudo-null experiments by splitting or resampling baseline trials into groups with the same sizes as the real conditions. Recompute INS and WTD in each pseudo-null experiment to estimate the mean sampling floor; subtracting that floor is a correction, not an unbiased recovery of latent instability. Reports should include the raw estimate, estimated floor, untruncated difference, and a bootstrap confidence interval clustered by item. This paper also recommends leaving negative adjusted values visible, since they convey correction uncertainty.

To avoid using the same noise for tail selection and estimation, use disjoint samples. One half selects the items with the smallest estimated changes; the other estimates their degradation. Reverse the roles and average over repeated random splits to use the data symmetrically. The held-out half estimates degradation for the noisy-selected set; noisy selection can still miss the oracle tail of latent changes.

The design fails or changes interpretation when baseline and perturbed item sets aren't paired, the alleged perturbation adds task-relevant information, outputs are deterministic one-shots, profiles

have zero variance, floor or ceiling effects dominate, or N_g is too small for the chosen tail. Results from heterogeneous perturbations and delays shouldn't be pooled merely to obtain a larger sample. Nor should one model's worst-tail item set be presumed to transfer without validation, given Zhang et al.'s evidence that affected items are largely model-specific.

3.6 WHY NO GENERAL ROBUSTNESS SCALAR

Profile similarity, signed change, item instability, and harmful-tail degradation aren't noisy measures of one latent quantity. They answer different questions. A high PSS with a strongly negative L means that the profile retained its shape while falling in level. A near-zero L with high INS means that changes cancelled. Moderate INS with high WTD means that instability is concentrated in a harmful tail.

The primary output is a dashboard indexed by domain and perturbation. A scalar is warranted only when a named target Y validates a function

$$f_Y(\text{PSS}, L, \text{INS}, \text{WTD}, \dots)$$

on held-out systems. The fitted function is then a target-specific predictive or decision model, not a universal index of general intelligence.

4 RETENTION, FEEDBACK, AND SCAFFOLD DEPENDENCE

4.1 DURABLE RETENTION

Retention is a longitudinal projection, not a component of generic robustness. Pre-register novel facts, procedures, or concepts; verify immediate performance; and retest at declared delays in clean sessions. Item screening for baseline floor and ceiling effects should use data independent of the data used to estimate retention. Otherwise selection on a noisy immediate score creates regression to the mean.

At each delay d , compute signed change, INS, and WTD exactly as in Section 3. Don't average delays by default, since a system that holds for 24 hours and fails after seven days has a meaningful trajectory.

For taught items indexed by t , if a one-sided retained-mass summary is useful, define

$$\text{DR}_d = \frac{\sum_t \min(s_{t0}, s_{td})}{\sum_t s_{t0}}, \quad (3)$$

provided the denominator is positive. This asks how much immediately demonstrated performance remains at delay d without allowing later gains to offset losses on other items. Gains remain visible in the signed item changes. The measure is behavioural: a high DR_d doesn't by itself show that the base model consolidated an internal memory.

4.2 RESPONSE TO FEEDBACK

A separate protocol asks whether error rates fall over repeated attempts with a declared form of feedback. Let $e_k \in [0, 1]$ be the error rate at attempt k , for $k = 1, \dots, K$ and $K \geq 2$. Report endpoint gain

$$G = e_1 - e_K \quad (4)$$

and define total transition variation and its backsliding component as

$$V = \sum_{k=2}^K |e_k - e_{k-1}|, \quad U = \sum_{k=2}^K (e_k - e_{k-1})_+.$$

The backsliding share is

$$B = \begin{cases} U/V, & V > 0, \\ 0, & V = 0. \end{cases} \quad (5)$$

Here $(x)_+ = \max(0, x)$. $G > 0$ indicates net improvement; lower $B \in [0, 1]$ indicates that less transition variation came from error increases. B isn't the fraction of attempts that got worse, and multiplying G and $1 - B$ would create a composite with no validated interpretation. Report the two quantities separately, with task-level trajectories and uncertainty.

Feedback type is part of the intervention. "Try again", self-review, tool feedback, and an explicit error diagnosis provide different information. Results transport only across feedback regimes for which transport has been tested.

4.3 THE UNIT OF EVALUATION AND SCAFFOLD DEPENDENCE

Context windows, retrieval systems, tools, and persistent stores complicate the bearer of a score. A base model and a deployed agent are different units. A retrieval system needn't be dismissed as a fake capability merely because the base model performs poorly without it; the relevant question is whether the projection concerns the base model or the deployed composite.

For a declared system, test at least three configurations:

1. the intended deployment configuration;
2. one or more degraded configurations, such as noisy retrieval, restricted context, or weaker tools;
3. a minimal configuration that isolates the base model where that contrast is decision-relevant.

Report level, item instability, and harmful-tail changes for every transition. A graceful curve supports a projection about component degradation within the tested range. A sharp drop supports a claim of component dependence. Neither result, by itself, determines which configuration is the "real" agent.

Construct validity remains separate. A multimodal system that performs visual question answering without images may be exploiting language priors rather than visual information (Asadi et al., 2026). A scaffold-degradation test can reveal dependence, but deciding what ability was measured requires interventions that distinguish rival routes to performance.

4.4 WHAT BEHAVIOUR CAN'T ESTABLISH

Yetman (2026) argues that behavioural testing alone can't establish with certainty whether a performance relies on a representation. Probes can test whether information is present and exploitable; activation patching, steering, or ablation can increase confidence that a representation plays a causal role. Interventions on identified sparse neuron sets that change agreement behaviour illustrate one possible route from behavioural regularity to mechanistic evidence (Arora et al., 2026).

These methods can strengthen an explanation of why a projection holds. They remain additional evidence. The dashboard in this paper is deliberately black-box and should be labelled as such. High retention isn't yet an internal consolidation mechanism; declining errors aren't yet corrective control; stable profiles aren't yet a homeostatic cluster.

5 VALIDATION AND WEIGHTING

5.1 INCREMENTAL PREDICTIVE VALIDITY

Let Y denote the declared external outcome: deployment task success, calibrated risk, retention at a specified delay, or another target not used to construct the evaluation scores. The first comparison is with simple baselines. For a ten-domain battery,

$$S_{\text{equal}} = \frac{1}{G} \sum_{g=1}^G a_{g0}$$

is a useful reference. Depending on the target, a single snapshot score, model size, or prior incident rate may be another.

A richer evaluation earns its added complexity only if it improves held-out prediction of Y . Compare nested models such as

$$Y \sim S_{\text{equal}}$$

and

$$Y \sim a_0 + \text{PSS} + L + \text{INS} + \text{WTD} + \text{DR} + G + B.$$

The appropriate comparison depends on the outcome: proper scoring rules for probabilistic events, calibration error for risk estimates, and out-of-sample loss or explained variance for continuous outcomes. Report uncertainty in the increment, not just the fitted coefficients.

Every predictor in the comparison should be constructed without using the observations that define Y . If Y is seven-day retention, for example, DR computed from those same seven-day responses has to be excluded; retention at an earlier delay or on an independent item sample may instead serve as a predictor.

Validation also requires an outcome that is diagnostically independent of the score's construction. If the same failures define WTD and serve as its prediction target, the exercise is circular. Deployment incidents, future task success, or performance under a later perturbation family provide stronger tests.

5.2 TARGET-SPECIFIC WEIGHTS

For a target Y , a candidate weighted score is

$$S_Y = \sum_{g=1}^G w_g^{(Y)} a_{g0}, \quad w_g^{(Y)} \geq 0, \quad \sum_g w_g^{(Y)} = 1.$$

Weights may be estimated with regularization, preregistered from an independent theory, or given a shrinkage prior informed by human CHC data. In every case, compare S_Y with equal weights on data not used to select the weights.

Use nested cross-validation when the same dataset supports both selection and assessment. Prefer external validation on later model versions or distinct model families. Report calibration, sensitivity to the candidate domain decomposition, and whether the weights transport. A weight on storage that improves prediction of seven-day retention need not improve prediction of immediate tool-use success.

Causal interventions can also inform weighting. Degrading one component and measuring downstream changes can test whether it's upstream for a particular outcome. But causal importance still doesn't yield a purpose-free weight. The intervention, target, and deployment mixture determine what the effect means for a decision.

5.3 CHC AS A DEFEASIBLE MEASUREMENT VOCABULARY

CHC offers a developed vocabulary for broad cognitive abilities and an existing multidomain battery structure. It's useful here as an organizing hypothesis, not as an ontology of machine intelligence. The human positive manifold, differential g -loadings, and hierarchy may or may not reproduce across artificial systems.

Before transporting a CHC-based score across model families, test whether the domain definitions and item relations remain comparable. A decomposition that predicts one external outcome but fragments on another may still be useful within its scope. If alternative decompositions predict the target better, the evaluation should change. The category earns its keep through projection, not inheritance from human psychometrics.

6 OPERATIONAL WORKFLOW

The complete workflow can be implemented without architectural access:

1. Register a projection card. State Y , the population, unit, intervention range, time horizon, acceptable uncertainty, and any tail constraint.
2. Fix the measurement design. Specify domains, paired items, scoring rubrics, perturbation families, tail fraction q , delays, feedback regimes, and configuration changes.
3. Plan repeated sampling. Choose repetitions from a precision or power target and publish the compute budget. If adaptive allocation is used, keep the adaptation data separate from final estimation.
4. Run paired conditions. Balance sample counts across baseline and perturbations, randomize order where possible, and retain item identifiers.
5. Estimate each cell. Report baseline level, L , INS, WTD, transition distributions, and confidence intervals for every domain-by-perturbation cell. Report PSS separately at profile level.
6. Correct nonlinear noise. Use baseline-only pseudo-null resampling for INS and WTD, and use disjoint samples for tail selection and estimation.
7. Validate externally. Fit any target-specific weighting or composite on development data and assess it on held-out systems and outcomes.
8. Release an auditable report. Publish seeds after testing, scoring code, raw and adjusted estimates, missingness, exclusions, compute use, and all preregistered deviations.

Lottery-style sampling from declared perturbation families can reduce direct test-item optimization. It doesn't make the results intrinsically valid or ungamable. The sampled conditions still need to represent the intervention range named in the projection card.

7 PREDICTIONS AND FAILURE CONDITIONS

The framework makes claims that can fail.

PI: Granular measures add information. For at least some deployment targets, item instability and harmful-tail degradation will improve held-out prediction beyond baseline accuracy, signed mean change, and profile similarity.

Failure condition: If INS and WTD add no predictive or decision value across adequately powered, externally validated studies, the extra sampling isn't justified for those targets.

P2: The useful granularity is target-dependent. Domain-profile stability will predict some broad outcomes, while item and tail measures will better predict outcomes driven by isolated failures. No granularity will dominate across all targets.

Failure condition: If one simple aggregate consistently matches or beats the dashboard across targets, populations, and interventions, the multi-granular design should be reduced to that aggregate.

P3: Weight transport is limited. Weights optimized for one target or model population will often lose calibration or predictive value under a different target, family, or deployment mixture.

Failure condition: If one stable weight vector repeatedly transports without loss across these changes, a general weighting scheme becomes empirically defensible.

P4: Behavioural and mechanistic evidence can dissociate. Systems with similar dashboards may rely on different internal routes, while systems with different internal implementations may support the same declared projection.

Failure condition: If dashboard similarity predicts preregistered mechanistic-similarity measures with near-ceiling held-out accuracy across model families, the evidential boundary drawn in Section 4.4 is too conservative.

These predictions are deliberately comparative. They don't assume universal numerical cut-offs before distributions and external outcomes exist.

8 DECISION USE AND GOVERNANCE

Governance needs decisions, not merely more scores. A regulator or release committee should specify the loss associated with false reassurance, false alarms, and harmful-tail events. The evaluation can then be calibrated against those losses.

No generic capability–stability tiers are justified at present. A threshold for monitoring, release, or independent review should be estimated from the relevant outcome data and accompanied by uncertainty. A stable mean can't override an unacceptable tail constraint; conversely, a noisy tail estimate with inadequate sample size shouldn't trigger a confident decision.

Reports should keep the base model and deployed system distinct, identify the tested configurations, and state where transport is untested. This improves governance even before a predictive composite is available. It prevents one attractive number from hiding the assumptions about population, unit, and intervention range on which the decision actually depends.

Multi-granular reporting can make narrow optimization more visible. Offsetting improvements and degradations, dependence on a scaffold, and failure concentrated in one domain are harder to hide when cells and tails are published. It doesn't make gaming impossible, nor does a high dashboard establish "genuine general intelligence".

9 LIMITATIONS AND EXTENSIONS

First, the framework inherits any construct-validity failures in its battery. Disaggregating an invalid test doesn't make it valid. CHC domains are a defeasible starting point, and comparable labels needn't have measurement invariance across biological and artificial systems.

Second, repeated item-level sampling is expensive. Tail estimates require enough items and enough repetitions to distinguish latent change from sampling noise. Adaptive allocation may

reduce cost, but it complicates inference and can create new selection bias.

Third, an invariance test is only as good as its perturbation. Task-irrelevant context is a clean case because it's designed not to change the answer. Distribution shifts, tool changes, and scaffold removal often change the task itself. Each protocol has to state which property is expected to remain invariant and why.

Fourth, worst-tail membership may be unstable across models and versions. This limits transport but can itself be decision-relevant. Evaluation should measure the target system rather than assume that another model's difficult items identify its risks.

Fifth, the framework is behavioural. It can establish predictive regularities without settling whether a system represents, consolidates, understands, or controls. Mechanistic interpretability may supply stronger evidence, but it introduces its own validity and transport problems.

Finally, the proposal doesn't settle the ontology or boundary of AGI. It asks a narrower question: what does an evaluation allow us to infer, and how do we know? That narrower target is enough to rule out several common overclaims and to improve present measurement practice.

Formal verification can establish that metric definitions are bounded under stated assumptions. It can't establish construct validity, sampling adequacy, predictive validity, or a homeostatic mechanism. The revised Lean companion formalizes only algebraic invariants of the retained estimands and leaves the statistical protocol as an empirical obligation.

10 CONCLUSION

Multidomain profiles are a necessary improvement over narrow benchmarks, but breadth alone doesn't make an AGI evaluation projectible. Aggregate accuracy can conceal item-level flips. Stable domain shape can coexist with falling levels and harmful tails. Repeated black-box performance can remain silent about mechanism.

The remedy is to design backward from the inference or decision. Declare the target, population, unit, intervention range, and time horizon. Report profile shape, signed level, item instability, and harmful-tail degradation separately. Treat retention, feedback response, and scaffold dependence as distinct protocols. Use equal weighting as a baseline and allow alternatives only when they improve held-out prediction and calibration.

This approach is deliberately less ontologically ambitious than a homeostatic account of general intelligence. It's also more demanding empirically. A metric earns its place by what it predicts beyond itself, at the granularity and over the range for which the claim is made.

A FORMAL SANITY CHECK OF THE ESTIMANDS

A **companion Lean 4 module** checks range properties of the retained estimands. For an admissible correlation $r \in [-1, 1]$, PSS lies in $[0, 1]$. With item scores in $[0, 1]$, signed mean change and WTD lie in $[-1, 1]$, while INS lies in $[0, 1]$. Retained mass additionally requires non-negative scores and a positive baseline total. Error rates in $[0, 1]$ bound endpoint gain to $[-1, 1]$; the exact backsliding share lies in $[0, 1]$, with a constant trajectory assigned $B = 0$.

The tail is an external selector, so the WTD proof checks only the range for a selected set, not whether a noisy ranking recovers the latent tail. Correlation estimation, pseudo-null correction, split-sample selection, and predictive or construct validity remain outside the formalization. Algebraic bounds establish neither reliability nor projectibility, and they don't identify a mechanism.

REFERENCES

- Arora, A., Wu, Z., Steinhardt, J., & Schwettmann, S. (2026). Language model circuits are sparse in the neuron basis.
- Asadi, M., O’Sullivan, J. W., Cao, F., Nedae, T., Fardi, K., Li, F.-F., Adeli, E., & Ashley, E. (2026). MIRAGE: The illusion of visual understanding.
- Boyd, R. (1991). Realism, anti-foundationalism and the enthusiasm for natural kinds. *Philosophical Studies*, 61(1–2), 127–148. <https://doi.org/10.1007/BF00385837>
- Carroll, J. B. (1993). *Human cognitive abilities: A survey of factor-analytic studies*. Cambridge University Press. <https://doi.org/10.1017/CBO9780511571312>
- Hendrycks, D., Song, D., Szegedy, C., Lee, H., Gal, Y., Li, S., Zou, A., Levine, L., Han, B., Fu, J., Liu, Z., Shin, J., Lee, K., Mazeika, M., Phan, L., Ingebreetsen, G., Khoja, A., Xie, C., Salaudeen, O., ... Bengio, Y. (2025). A Definition of AGI [v3, December 2025].
- Hernández-Orallo, J. (2017). *The measure of all minds: Evaluating natural and artificial intelligence*. Cambridge University Press. <https://doi.org/10.1017/9781316594179>
- Kuribayashi, T., Warstadt, A., Oseki, Y., & Wilcox, E. G. (2026). Dual alignment between language model layers and human sentence processing. *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 46207–46223. <https://doi.org/10.18653/v1/2026.acl-long.2143>
- McGrew, K. S. (2009). CHC theory and the human cognitive abilities project: Standing on the shoulders of the giants of psychometric intelligence research. *Intelligence*, 37(1), 1–10. <https://doi.org/10.1016/j.intell.2008.08.006>
- Reynolds, B. (2026). *Not every stable cluster is homeostatic: Stability, network order, and control in projectible kinds* [Manuscript archived at PhilArchive, 1 June 2026]. <https://philarchive.org/rec/REYNES>
- Schneider, W. J., & McGrew, K. S. (2018). The Cattell-Horn-Carroll theory of cognitive abilities. In D. P. Flanagan & E. M. McDonough (Eds.), *Contemporary intellectual assessment: Theories, tests, and issues* (4th, pp. 73–163). The Guilford Press.
- von der Malsburg, T., & Padó, S. (2026, March 17). *Diverging transformer predictions for human sentence processing: A comprehensive analysis of agreement attraction effects* [arXiv preprint].
- Weinberger, N. (2026). Homeostasis and causal control. *Biology & Philosophy*, 41, Article 18. <https://doi.org/10.1007/s10539-026-10018-8>
- Yetman, C. (2026). Representation in large language models [Forthcoming]. *Ergo: An Open Access Journal of Philosophy*. <https://doi.org/10.3998/ergo.10324>
- Zhang, Y., Koyejo, S., & Yang, D. (2026, July 15). *The illusion of robustness: Aggregate accuracy hides prediction flips under task-irrelevant context* [arXiv preprint, version 2].